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Hana Network

SECURITY REVIEW

Date: 28 May 2024

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1. About Shieldify

Positioned as the first hybrid Web3 Security company, Shieldify shakes things up with a unique subscription-based auditing model that entitles the customer to unlimited audits within its duration, as well as top-notch service quality thanks to a disruptive 6-layered security approach. The company works with very well-established researchers in the space and have secured multiple millions in TVL across protocols, also can audit codebases written in Solidity, Vyper, Rust, Cairo, Move and Go.

Learn more about us at shieldify.org.

2. Disclaimer

This security review does not guarantee bulletproof protection against a hack or exploit. Smart contracts are a novel technological feat with many known and unknown risks. The protocol, which this report is intended for, indemnifies Shieldify Security against any responsibility for any misbehavior, bugs, or exploits affecting the audited code during any part of the project's life cycle. It is also pivotal to acknowledge that modifications made to the audited code, including fixes for the issues described in this report, may introduce new problems and necessitate additional auditing.

3. About Hana Network

Hana is the first privacy layer between Ethereum and Bitcoin, complying with regulations, and enabling on-chain privacy on existing chains and for arbitrary assets. We are selected for Binance Labs Incubation and won the 2nd prize at the Zkday pitch competition in 2023.

Hana Chain is an EVM-compatible L1 blockchain that enables omnichain, generic smart contracts and messaging between any blockchain.

Prerequisites

- Go 1.20
- Docker and Docker Compose (optional, for running tests locally)
- buf (optional, for processing protocol buffer files)
- jq (optional, for running scripts)

Components of Hana Chain

Hana Chain is built with Cosmos SDK, a modular framework for building blockchain and Ethermint, a module that implements EVM-compatibility.

- hana-node (this repository) contains the source code for the HanaChain node (**hanacored**) and the HanaChain client (**hanaclientd**).
- protocol-contracts contains the source code for the Solidity smart contracts that implement the core functionality of HanaChain.

4. Risk Classification

Severity	Impact: High	Impact: Medium	Impact: Low
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

5. Security Review Summary

The security review lasted 1 week with a total of 224 hours dedicated to the audit by four researchers from the Shieldify team.

Overall, the code is well-written and leverages Zetachain's codebase as a foundation. The audit report contributed by identifying two issues of low severity, covering the usage of whitelist approach, as well as using Cosmos SDK and Go versions that are prone to vulnerabilities.

We would like to extend our gratitude to the Hana Network team for their fast and meticulous responses.

5.1 Protocol Summary

Project Name	**Hana Network**
Repository	hana-node
Type of Project	EVM-compatible L1 blockchain, Cosmos SDK, Go 1.20
Audit Timeline	5 days
Review Commit Hash	273dea4c1e73025501c8b8c0f9c7713c4d152be2

5.2 Scope

The following smart contracts were in the scope of the security review:

app/ante/ante.go
app/ante/ante_test.go
app/ante/handler_options.go
app/app.go
changelog.md
localnet/scripts/start-hanacored.sh
seeding-devnet/README.md
seeding-devnet/scripts/start-hanacored-1.sh

seeding-devnet/scripts/start-hanacored-2.sh
seeding-devnet/scripts/start-hanacored-for-generating-node.sh
seeding-devnet/scripts/start-hanacored-for-sync-node.sh

6. Findings Summary

The following number of issues have been identified, sorted by their severity:

- **Low** issues: **2**

ID	Title	Severity	Status
[L-01]	Consider Using the Whitelist Approach Over the Blacklist	Low	To be fixed
[L-02]	Vulnerable Cosmos SDK and Go-Ethereum Version Used	Low	To be fixed

7. Findings

[L-01] Consider Using the Whitelist Approach Over the Blacklist

Severity

Low Risk

Description

The `blockedReceivingModAcc` mapping configures module accounts that cannot receive tokens. This approach means that future-added modules can receive tokens from users by default, which may be problematic if the modules are not expected to receive tokens and cannot be sent out.

Location of Affected Code

File: app/app.go#L208-L214

```
blockedReceivingModAcc = map[string]bool{
    distrtypes.ModuleName:          true,
    authtypes.FeeCollectorName:     true,
    stakingtypes.BondedPoolName:    true,
    stakingtypes.NotBondedPoolName: true,
    govtypes.ModuleName:            true,
}
```

Recommendation

Consider implementing a whitelist approach of what modules can receive tokens over a blacklist module with an `allowedReceivingModAcc` mapping.

Team Response

To be fixed.

[L-02] Vulnerable Cosmos SDK and Go-Ethereum Version Used

Severity

Low Risk

Description

The codebase uses a `cosmos-sdk` version of `v0.46.13`, vulnerable to the "ASA-2024-005: Potential slashing evasion during re-delegation" issue.

Additionally, the `go-ethereum` version used is `v1.10.26`, which is vulnerable to denial of service via malicious p2p messages, as indicated in GHSA-ppjg-v974-84cm and GHSA-4xc9-8hmq-j652.

Location of Affected Code

File: go.mod#L6-L7

```
require (
    github.com/cosmos/cosmos-sdk v0.46.13
    github.com/ethereum/go-ethereum v1.10.26
```

Recommendation

Consider updating `cosmos-sdk` and `go-ethereum` to their latest stable version.

Team Response

To be fixed.

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Thank you!